

SOLVING 2ND O.D.Es

Lesson 25

* Method of undetermined coefficients, can be applied:

Step 1: Find Complementary soln; $y_c(x)$ {Based on a homogeneous d.e}

Step 2: Assume Particular soln; $y_p(x)$ {Based on $f(x)$ }

Step 3: General soln:

$$y(x) = y_c(x) + y_p(x)$$

Example 1

Solve (a) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 5y = 13e^{3x}$

(b) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 8y = 2xe^x$

Solution:

(a) CS: $m^2 + 4m + 5 = 0$

$$m = \frac{-4 \pm \sqrt{16 - 4(1)(5)}}{2} = \frac{-4 \pm \sqrt{-4}}{2} = -2 \pm i \quad (\text{Complex roots})$$

$$\therefore y_c(x) = e^{-2x} (A \cos x + B \sin x)$$

PS: Assume $y_p = ae^{3x}$, Since $f(x) = 13e^{3x}$

then $y'_p = 3ae^{3x}$; $y''_p = 9ae^{3x}$

Substitute:

$$9ae^{3x} + 4(3ae^{3x}) + 5ae^{3x} = 13e^{3x}$$

$$26ae^{3x} = 13e^{3x}$$

$$26a = 13, \therefore a = \frac{1}{2}$$

$$y_p(x) = \frac{1}{2}e^{3x}$$

General

Soln:

$$y(x) = y_c(x) + y_p(x)$$

$$y(x) = e^{-2x} (A \cos x + B \sin x) + \frac{1}{2}e^{3x}$$

(b) Aux eqn: $m^2 - 2m - 8 = 0$

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$$m = \frac{2 \pm \sqrt{4 - 4(1)(-8)}}{2} = \frac{2 \pm \sqrt{36}}{2}$$

(Real & distinct)

C.S:

$$m = \frac{2+6}{2} = 4 \quad \text{or} \quad m = \frac{2-6}{2} = -2$$

Complementary
Soln

$$y_c = Ae^{4x} + Be^{-2x}$$

P.S: Assume particular Soln

$$y_p = (ax+b)e^{-x}; \quad \text{Since } f(x) = 2xe^{-x}$$

$$y'_p = ae^{-x} - (ax+b)e^{-x}$$

$$y''_p = -ae^{-x} + (ax+b)e^{-x} - ae^{-x} = -2ae^{-x} + (ax+b)e^{-x}$$

Substitute:

$$-2ae^{-x} + (ax+b)e^{-x} - 2(ae^{-x} - (ax+b)e^{-x}) - 8(ax+b)e^{-x} = 2xe^{-x}$$

$$-2ae^{-x} + (ax+b)e^{-x} - 2ae^{-x} + 2(ax+b)e^{-x} - 8(ax+b)e^{-x} = 2xe^{-x}$$

$$-5ax e^{-x} + (-4a-5b)e^{-x} = 2x e^{-x}$$

Compare

Coefficients:

$$-5a = 2; \quad -4a - 5b = 0$$

$$\therefore a = -\frac{2}{5}$$

$$+\frac{8}{5} - 5b = 0, \quad \therefore b = \frac{8}{25}$$

P.S

$$y_p = \left(-\frac{2}{5}x + \frac{8}{25}\right)e^{-x}$$

General Soln:

$$y(x) = y_c + y_p$$

$$\therefore y(x) = Ae^{4x} + Be^{-2x} + \left(\frac{8}{25} - \frac{2}{5}x\right)e^{-x}$$

Example 2: <Be careful>

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Solve: (a) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 8y = 3e^{-2x}$

(b) $\frac{d^2y}{dx^2} + y = 2\cos x$

Solution

(a) C.S; $m^2 - 2m - 8 = 0$ ie $(m-4)(m+2) = 0$
 $\underline{m=4}$ & $\underline{m=-2}$ (Real & distinct roots)

$y(x) = Ae^{4x} + Be^{-2x}$

P.S Assume $y_p = axe^{-2x}$, since ae^{-2x} is already in C.S

$y'_p = ae^{-2x} - 2axe^{-2x}$

$y''_p = -2ae^{-2x} + 4axe^{-2x} - 2ae^{-2x} = -4ae^{-2x} + 4axe^{-2x}$

Substitute:

$-4ae^{-2x} + 4axe^{-2x} - 2(ae^{-2x} - 2axe^{-2x}) - 8axe^{-2x} = 3e^{-2x}$

$-4ae^{-2x} + 4axe^{-2x} - 2ae^{-2x} + 4axe^{-2x} - 8axe^{-2x} = 3e^{-2x}$

$6ae^{-2x} = 3e^{-2x}$ ie $\begin{cases} 6a = 3 \\ \therefore a = \frac{1}{2} \end{cases}$

P.S:

$y_p(x) = \frac{1}{2}xe^{-2x}$

General

Soln

$y(x) = y_c(x) + y_p(x)$

$y(x) = Ae^{4x} + Be^{-2x} + \frac{1}{2}xe^{-2x}$

(b) CS: $m^2 + 1 = 0$, $m = \pm\sqrt{-1} = \pm i$ (Complex roots) -4-

Complementary
Solu

$$y_c(x) = A \cos x + B \sin x \quad \checkmark$$

PS, Assume $y_p = ax \cos x + bx \sin x$, since $a \cos x + b \sin x$ - already C.S.

$$y'_p = a \cos x + b \sin x - ax \sin x + bx \cos x$$

$$y''_p = -a \sin x + b \cos x - a \sin x + b \cos x - ax \cos x - bx \sin x$$

$$y''_p = -2a \sin x + 2b \cos x - ax \cos x - bx \sin x$$

Substitute:

$$-2a \sin x + 2b \cos x - \cancel{ax \cos x} - \cancel{bx \sin x} + \cancel{ax \cos x} + \cancel{bx \sin x} = 2 \cos x$$

$$-2a \sin x + 2b \cos x = 2 \cos x$$

Comparing Coefficients: $\begin{cases} a = 0 \\ b = 1 \end{cases}$

$$\therefore y_p(x) = x \sin x \quad \checkmark$$

General

Solu: $y(x) = y_c(x) + y_p(x)$

$$y(x) = A \cos x + B \sin x + x \sin x$$

Ex#3: Solve: (a) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = x^2 + 3e^{-2x}$

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(b) $\frac{d^2y}{dx^2} - 9y = 2e^{3x} + 3e^{2x}$

Solution

(a) CS: $m^2 + 4m + 4 = 0$; $(m+2)^2 = 0$
 $m = -2$ (Real & equal roots)

$y_c = Ae^{-2x} + Bxe^{-2x}$

PS: (There two particular solutions):

Assume $y_p = ax^2 + bx + c$; $y'_p = 2ax + b$; $y''_p = 2a$

Substitute: $2a + 4(2ax + b) + 4(ax^2 + bx + c) = x^2$
 $4ax^2 + (4b + 8a)x + 2a + 4b + 4c = x^2$

$4a^2 = 1$

$4b + 8a = 0$

$2a + 4b + c = 0$

$\therefore a = \frac{1}{4}$

$4b + 2 = 0$

$\frac{1}{2} - 2 + c = 0$

$\therefore b = -\frac{1}{2}$

$\therefore c = \frac{3}{2}$

$\therefore y_{p_1} = \frac{1}{4}x^2 - \frac{1}{2}x + \frac{3}{2}$

Assume, $y_{p_2} = dx^2e^{-2x}$; $y'_{p_2} = 2dx e^{-2x} - 2dx^2e^{-2x}$

$y''_{p_2} = 2de^{-2x} - 4dx e^{-2x} - 4dx e^{-2x} + 4dx^2e^{-2x} = 2de^{-2x} - 8dx e^{-2x} + 4dx^2e^{-2x}$

Substitute:

$2de^{-2x} - 8dx e^{-2x} + 4dx^2e^{-2x} + 4(2dx e^{-2x} - 2dx^2e^{-2x}) + 4dx^2e^{-2x} = 3e^{-2x}$
 $2de^{-3x} = 3e^{-2x}$

$y_{p_2} = \frac{3}{2}x^2e^{-2x}$

$\therefore d = \frac{3}{2}$

General Soln: $y(x) = y_c + y_{p_1} + y_{p_2}$

$y(x) = Ae^{-2x} + Be^{-2x} + \frac{1}{4}x^2 - \frac{1}{2}x + \frac{3}{2} + \frac{3}{2}x^2e^{-2x}$

(b) CS: $m^2 - 9 = 0$; $m = \pm 3$ (Real & distinct roots)

$$y_c = Ae^{3x} + Be^{-3x}$$

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PS: (There are two particular solutions)

Assume $y_{P_1} = axe^{3x}$, (since Ae^{3x} is already a solⁿ)

$$y'_{P_1} = ae^{3x} + 3axe^{3x}$$

$$y''_{P_1} = 3ae^{3x} + 3ae^{3x} + 9axe^{3x} = 6ae^{3x} + 9axe^{3x}$$

Substitute: $6ae^{3x} + 9axe^{3x} - 9axe^{3x} = 2e^{3x}$

$$6ae^{3x} = 2e^{3x}$$

$$\therefore a = \frac{1}{3}$$

$$y_{P_1}(x) = \frac{1}{3}xe^{3x}$$

Assume, $y_{P_2} = be^{2x}$, $y'_{P_2} = 2be^{2x}$; $y''_{P_2} = 4be^{2x}$

Substitute: $4be^{2x} - 9be^{2x} = 3e^{2x}$

$$-5be^{2x} = 3e^{2x}$$

$$\text{ie } b = -\frac{3}{5}$$

$$y_{P_2}(x) = -\frac{3}{5}e^{2x}$$

General Solution: $y(x) = y_c + y_{P_1} + y_{P_2}$

$$y(x) = Ae^{3x} + Be^{-3x} + \frac{1}{3}xe^{3x} - \frac{3}{5}e^{2x}$$